RMO(INDIA) – 2011

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Let ABC be a triangle. Let D, E, F be points respectively on the segments BC, CA, AB such that AD, BE, CF concur at point K. Suppose $\frac{BD}{DC} = \frac{BF}{FA}$ and $\angle ADB = \angle AFC$. Prove that $\angle ABE = \angle CAD$.
2. Let $(a_1, a_2, ..., a_{2011})$ be a permutation (arrangements) of the numbers 1, 2, 3, …., 2011. Show that there exist two number j, k such that $1 \le j \le k \le 2011$ and $\left|a_j - j\right| = \left|a_k - k\right|$.
3. A natural number n is chosen strictly between two consecutive perfect squares. The smaller of these two squares is obtained by subtracting k from n and the larger one is obtained by adding l to n. Prove that $n - kl$ is a perfect square.
4. Consider a 20-sided convex polygon K, with integers $A_1, A_2, ..., A_{20}$ in that order. Find the number of ways in which three sides of K can be chosen that every pair among them has at least two sides of K between them. For example $(A_1A_2, A_4A_5, A_{11}A_{12})$ admissible triple while $(A_1A_2, A_4A_5, A_{19}A_{20})$ is not.
5. Let ABC be a triangle and let BB_1, CC_1 be respectively the bisector of $\angle B, \angle C$ with B_1 on AC and C_1 on AB. Let E, F be the feet of perpendiculars drawn from A onto BB_1 and CC_1 respectively. Suppose D is the point at which the incircle of ABC touches AB. Prove that $AD = EF$.
6. Find all pairs of (x, y) of real numbers such that $16^{x^2+y} + 16^{x+y^2} = 1$.

******
****
**